Project: Causal Machine Learning

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Throughout the project 5-fold cross-validation and cross-fitting are used in all methods. When using SuperLearner, the following methods are included: generalized linear models (SL.glm), Random Forest (SL.ranger), generalized additive models (SL.gam) and penalized regression using elastic net (SL.glmnet). The division of the project was as follows: Sarah covered questions 1-3, Charlotte question 4 and Eline question 5.

1 Question 1

This project focuses on analyzing the effect of participation in 401(k) plans on accumulated assets. Since participation is not randomly assigned and may be confounded by individuals' saving preferences, using *eligibility* for a 401(k) plan, i.e. whether the employer offers such a plan, as the exposure variable is more appropriate than using actual participation. At the time the data was collected, it was unlikely that individuals chose their jobs based on the availability of a 401(k) plan. As a result, saving preferences likely did not influence job choice and thus did not directly affect eligibility status. The primary potential confounder in the relationship between eligibility and accumulated assets is *income*. Higher-paying jobs are generally more likely to offer 401(k) plans, making individuals with higher incomes more likely to be eligible. Moreover, individuals with higher incomes often have stronger saving preferences, which can lead to higher asset accumulation. Based on this reasoning, we assume that eligibility for a 401(k) plan is *conditionally exchangeable given income*, denoted as $E \perp\!\!\!\perp A^e \mid I$. To assess this assumption, we refer to the SWIG (Single World Intervention Graph) shown in Figure 1. If saving preferences did affect eligibility, a backdoor path would exist through the variable S, violating the exchangeability assumption. However, under the assumption that saving preferences do not influence eligibility, the only path connecting eligibility E and accumulated assets A runs through the confounder income. Conditioning on income blocks this path. Assuming that all relevant confounders are depicted in the diagram, the assumption of exchangeability appears plausible.

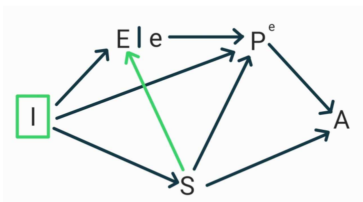


Figure 1: SWIG with variables participation (P), accumulated assets (A), preference for saving (S), eligibility (E) and income (I). The green line shows a possible effect of preference for saving on eligibility, which is assumed to be zero.

2 Question 2

As it can be interesting to know the effect of eligibility on net financial assets for a random household in our study as well as the effect among the households eligible for enrolling in a 401(k) plan, we will report both the Average Treatment Effect (ATE) and the Average Treatment Effect among the Treated (ATT).

Using the R-packages `npcausal` and `tmle` we will estimate the ATE and ATT using Augmented Inverse Probability Weighting (AIPW) and Targeted Maximum Likelihood Estimation (TMLE). The covariates that were used are the following: age, income, family size, education (in years), married, two earners, defined benefit pension, individual retirement account and home owner. This results in an ATE of 8190 with AIPW and 8297 with TMLE. This we interpret as follows; suppose all of the observed people were eligible for a 401(k) plan, then the average net financial assets among people with the same values in the other covariates would be approximately 8000 dollars more than if no one was eligible for a 401(k). Both these results are statistically different from zero as is seen in the confidence intervals in Table 1.

Method	ATE	ATE lb	ATE up	ATT	ATT lb	ATT up
AIPW	8190	6020	10361	10695	8147	13243
AIPW (no cov)	19543	16778	22309	19559	16789	22328
TMLE	8297	6436	10159	56703	53941	59465
TMLE (no cov)	19545	16775	22315	19295	16526	22064

Table 1: Estimated values and their 95%-CI lower and upper bounds (lb and up) with and without (no cov) covariate adjustment.

The ATT is estimated at 10695 with AIPW and 56703 with TMLE. This we interpret as follows; on average a person that is eligible for a 401(k) plan has approximately 10700 (56700 resp.) dollars more net financial assets than if that person wouldn't have been eligible for a 401(k) plan (preserving the values of the other covariates). Again, the results are statistically different from zero with confidence intervals as in Table 1.

Both these methods make use of the positivity assumption. This assumption seems plausible since the propensity scores do not take on extreme values as is seen in Figures 5 and 6 in the appendix.

We compare these results with the estimated ATE and ATT using no covariate adjustment. With AIPW (TMLE resp.) this results in an ATE of 19543 (19545 resp.) dollars and an ATT of 19559 (19295) resp.) dollars. The corresponding confidence intervals can be found in Table 1. All of these values are significantly higher than with covariate adjustment. This means that without covariate adjustment the effect of eligibility on financial assets is overestimated.

3 Question 3

We estimate $\theta = E\{\text{Cov}(A, Y|L)\}$ by calculating its influence function and setting the mean to zero. In doing so, machine learning is used to estimate $E(A|L)$ and $E(Y|L)$. We obtain a value of 1480 with standard error 272 and confidence interval $[946, 2013]$. The p-value of the test of the null hypothesis that θ equals zero amounts to $5 * 10^{-08}$, therefore, the null hypothesis is rejected and we decide θ is statistically different from zero. We conclude that eligibility for enrolling in a 401(k) plan affects net financial assets.

4 Question 4

We would like to estimate the effect of eligibility for enrolling in a 401(k) plan on net financial assets within strata of the population. Using the R-learner, we estimate the CATE for individuals with the same age, income and years of education received. To obtain estimates for the propensity scores and the outcome based on the covariates, we should include all confounders listed in Question 2 in the covariates. Using cross-fitting and SuperLearner, we fitted these models and obtained estimates $\hat{E}(A_i|L_i)$ and $\hat{E}(Y_i|L_i)$. These estimates can then be used as inputs $\hat{p}$ and $\hat{m}$, respectively, in the R-learner. In the appendix in Figure 7 the estimated propensity scores are shown. All propensity scores lie between 0 and 1, which ensures the weights in the R learner are nonzero. There is also sufficient overlap between the eligible and ineligible groups.

With rboost we estimated the effect of eligibility for enrolling in a 401(k) plan on net financial assets in function of age, income and years of education. In Figure 2 a histogram is shown of the obtained effect estimates for all individuals in the study. The effect of being eligible for a 401(k) plan is estimated as positive for almost all people in the study. This means that, individuals with a given age, income, and education, are predicted to have more net financial assets if they would be eligible for a 401(k) plan, than if they would not be. Most often, it is estimated that an individual would have between $700 and $800 more net financial assets if they were eligible, than if they were not, given that their age, income and education is unchanged.

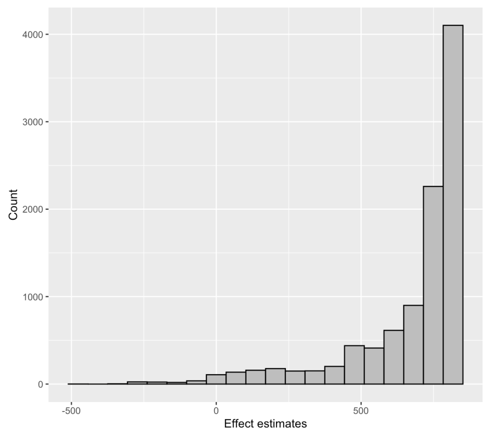
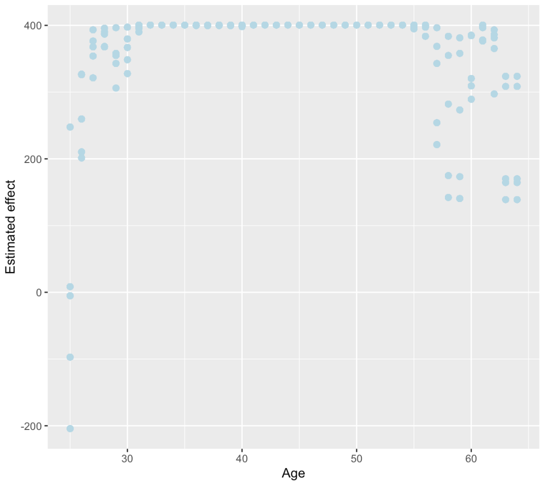


Figure 2: Histogram of R-learner estimates of effect of eligibility on net financial assets, in function of age, income and years of education.

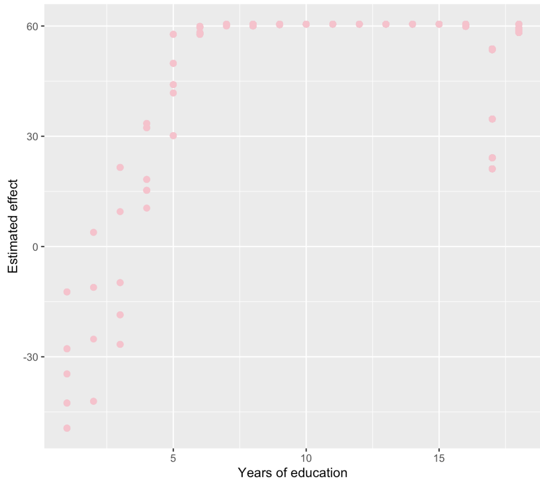
In addition, we visualize how the effect estimates vary by age, income and years of education received, separately. We obtained effect estimates using the same estimates for the outcomes and propensity scores as before, but now using only age or income or years of education as input feature in rboost.

In Figure 3a we see that the effect estimates vary most for people under the age of 30. It appears that for people under 30 the effect of eligibility for a 401(k) on net financial assets could be positive or negative, when we only stratify on their age. For people between the ages of 30 and 55, the estimated effect is always positive, with a value of around $400. Thus, an individual in that age bracket would be estimated to have $400 more net financial assets if they were elgible for a 401(k), than if they were not. Moreover, becoming a year older does not influence the magnitude of this effect, as long as they are younger than 55. For people above the age of 55 there is a bit more variation in the estimates again, but the effect of being eligible for a 401(k) plan on net financial assets remains positive. A possible explanation for this variation could be that people above the age of 55 start working less, and therefore are earning less and have less financial assets.

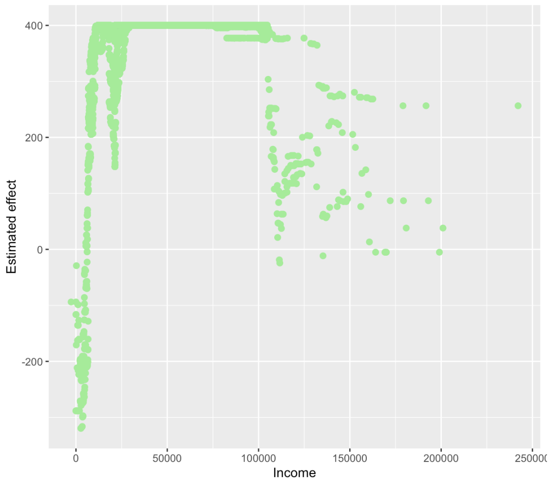
In Figure 3b it is shown how the effect estimates vary by years of education received. We see that for people who have received less than 5 years of education, the estimated effect of being eligible for a 401(k) plan on net financial assets can be negative. This would imply that people who have received little



(a) Estimated effects in function of age



(b) Estimated effects in function of education



(c) Estimated effects in function of income

Figure 3

education would be estimated to have less net financial assets if they were eligible for a 401(k), than if they were not. However, there are also individuals with little education for whom the estimated effect of eligibility is positive. The estimated effect of eligibility on net financial assets is highest for people who have received between 6 and 16 years of education. Such an individual would be estimated to have \$60 more net financial assets if they were eligible for a 401(k), than if they were not. For people who have had more than 16 years of education, the estimates are lower. Overall, we notice in the magnitude of the estimated effects of eligibility on net financial assets in function of education, compared to the estimated effects in function of age or income, that education seems to be less influential compared to age and income.

Figure 3c shows how the estimated effects vary by income. For people with the lowest incomes, the estimated effect of eligibility on net financial assets is negative. This would imply that someone with a very low income would have less net financial assets if they were eligible for a 401(k), than if they were not. It is likely that there are few data points (individuals) in this income region that are eligible for a 401(k) plan, so the estimates in this region may be inaccurate. Being eligible for a 401(k) plan seems to be most beneficial for people with incomes between \$30.000 and \$75.000. It is estimated that an individual with this income would have \$400 more net financial assets if they were eligible, than if they were not. Interestingly, for people with high incomes, eligibility has less of a positive influence on net financial assets.

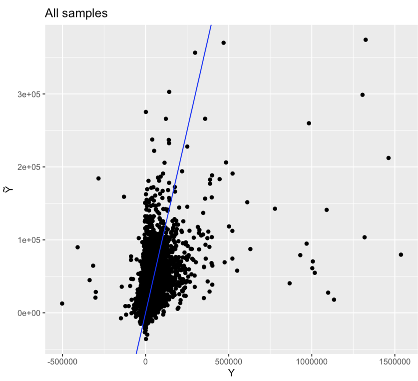
5 Question 5

The goal of this analysis is to estimate the net financial assets in the study population under the hypothetical scenario in which everyone is eligible to enroll in a 401(k) plan. This counterfactual outcome is denoted by Y^1. To estimate Y^1, a Doubly Robust (DR) learner is employed using the following pseudo-outcome:

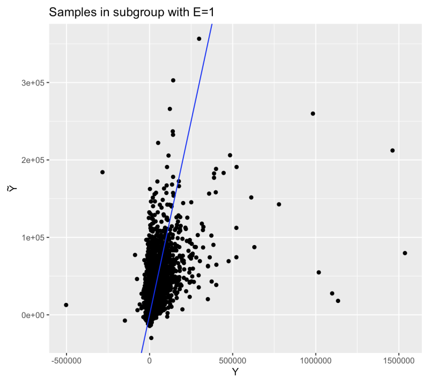
$$C_i = \hat{Y}_i^1 + \frac{E_i}{\hat{p}_i}(Y_i - \hat{Y}_i^1).$$

To predict the counterfactuals, 5-fold cross-fitting is applied twice in the DR-learner: once for constructing the pseudo-outcomes C_i, and once for estimating the final predictions $\tilde{Y}_i$. The propensity scores $\hat{p}_i$ and outcome predictions $\hat{Y}_i^1$ are estimated while controlling for the same set of covariates as in Question 2, to adjust for confounding. For predicting the pseudo-outcomes C_i, however, only age, income, and years of education are used as predictors.

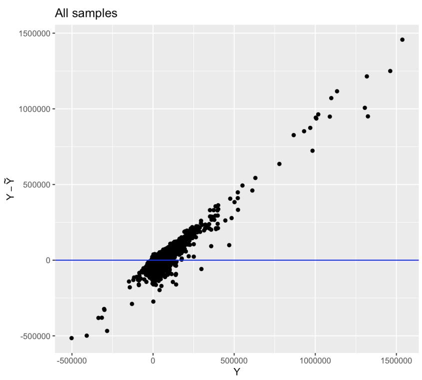
The resulting plots display: (a) the predicted counterfactual net financial assets $\tilde{Y}_i$ against the observed net financial assets Y_i for the entire population, (b) the same comparison restricted to individuals who were actually eligible for a 401(k) plan, and (c)-(d) the residuals (i.e. the difference between actual outcomes and predictions) plotted against the observed outcomes in both scenarios.



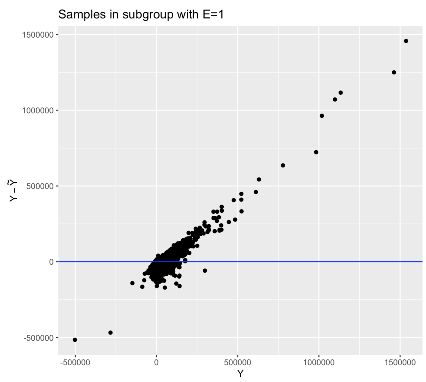
(a) Y vs. $\tilde{Y}$



(b) Y vs. $\tilde{Y}$ in subgroup with $E = 1$



(c) Difference between Y and $\tilde{Y}$ vs. Y



(d) Difference between Y and $\tilde{Y}$ vs. Y in subgroup with $E = 1$

Figure 4: Predicted counterfactual outcomes $\tilde{Y}^1$ versus observed outcomes Y, shown for the full sample and for the subgroup of individuals eligible for a 401(k) plan ($E = 1$). In the top row, the blue line represents the identity line $y = x$. In the bottom row, the blue line represents the zero-residual line $y = 0$.

Since the predictions are intended to estimate the counterfactual Y^1, relatively accurate estimates are expected within the subgroup of individuals who were actually eligible for a 401(k) plan ($E = 1$). As shown in Figure 4b, the predicted values are centered around the identity line $y = x$, which is an indication of good predicition within this subgroup. However, Figure 4d reveals that for more extreme values of Y (greater than 500.000 or less than -250.000, the predictions are noticeably worse. This suggests that the model tends to shrink extreme values toward the mean, underestimating both high and low net financial assets.

To assess the overall performance of the predictions, the mean squared error (MSE) of the counterfactual outcomes can be considered:

$$E\left[(Y^1 - \tilde{Y}^1)^2\right].$$

Because Y^1 is unobserved for individuals who were not eligible, this counterfactual MSE must be estimated. We use Augmented Inverse Probability Weighting (AIPW), with squared residuals $(Y - \tilde{Y})^2$ as outcomes, adjusting for the same set of covariates as in Question 2. The resulting estimate for the counterfactual MSE is:

$$\hat{E}\left[(Y^1 - \tilde{Y}^1)^2\right] = 3.150.831.174 \quad \text{with a 95\% confidence interval of } (1.892.329.275; 4.409.333.073).$$

While this estimated MSE is large, it does not necessarily imply that the model performs poorly across all samples. As observed in Figure 4d, the largest discrepancies between actual and predicted outcomes occur for extreme values of Y. Since the MSE is sensitive to outliers and squares these large differences, the high value is likely driven by a small number of extreme cases being underestimated, rather than indicating an overall poor prediction.

6 Appendix

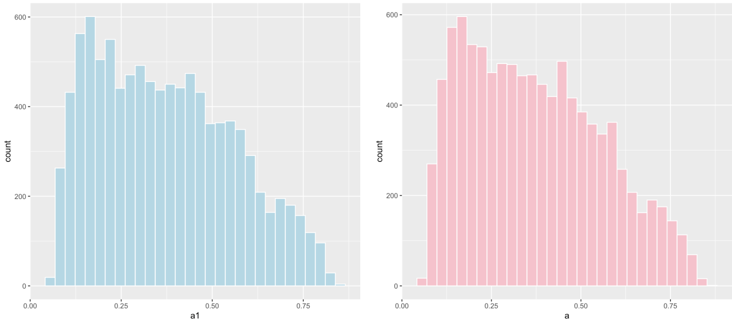


Figure 5: Propensity scores for the ATE (left) and the ATT (right) using AIPW in Question 2.

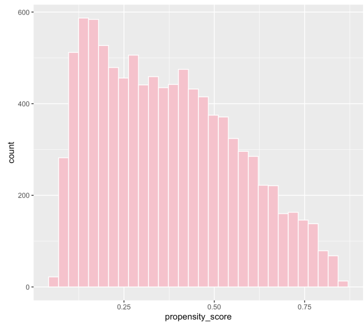


Figure 6: Propensity scores using TMLE in Question 2.

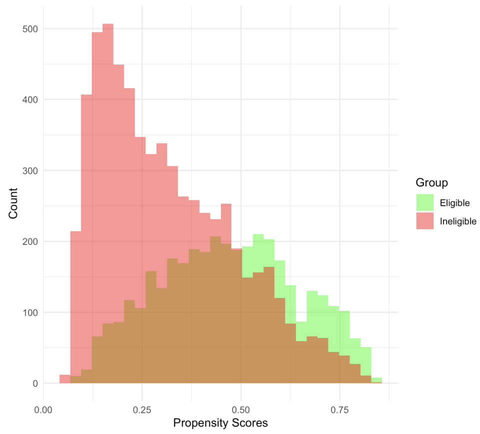


Figure 7: Estimated propensity scores used as input for the R learner in Question 4.